

Multiple Sclerosis: A Visual Perspective on Epidemiological Data

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Abstract—Multiple Sclerosis (MS) is a chronic, auto-immune demyelinating disorder of the central nervous system and constitutes one of the leading causes of non-traumatic disability in young adults aged 20–40. Over the past ten years this disease incidence has risen in every WHO region (i.e. stands for "World Health Organization" regions that are African, Americas, European, Western Pacific, South-East Asia, Eastern Mediterranean macro regions); the latest *Atlas of MS* estimates that more than 2.8 million people now live with the disease worldwide.

I have developed a web-based Visual Analytics application that unifies heterogeneous MS datasets (containing infos about epidemiology, population demographics, age onset and other info gathered from another dataset informations about lifestyle factors such as smoking prevalence) into a single, interactive solution allowing final users to have a tool for exploring data about this disease, user-friendly and easy to use.

I. INTRODUCTION

My main goal is to build an interactive Visual Analytics web application that transforms heterogeneous and static epidemiological data on Multiple Sclerosis (MS) into explorable tool that allows users to acquire a complete and deep knowledge about this disease.

Using the `d3.js` library at the front end and a Python backend implemented using Flask framework, the web app allows end users (i.e. researchers, medical specialists, neurologists, statisticians, epidemiologists, pharmaceutical analysts,...) to extract epidemiological details from a continental perspective down to state-level granularity.

A global choropleth map offers an immediate overview of MS cases distribution over the World and, using a graduated color scale, it's possible to instantly highlight the countries exhibiting the most pronounced trends for the selected metric. A stacked bar chart analyzes in details Multiple Sclerosis disease course (RRMS, SPMS, PPMS, Unknown) within macro-regions specified by user such that regional phenotypic differences are made more explicit and highlighted. To capture temporal dynamics, I added a parallel-coordinates chart where each line traces a region's trajectory across the three reference years (2008, 2013, 2020–22) and multiple epidemiological indicators.

I want also to focus on an analysis merging multiple aspects of MS and this multivariate relationship is represented using a bubble chart: the x-axis encodes number of MS cases, the y-axis number of smokers, and bubble size reflects the ratio between MS cases and smokers, allowing rapid identification of macro-regions exhibiting both high smoker cases and elevated number of MS diagnosis and enabling the

exploration of potential correlations or anomalies across broad geographic regions.

As last visualization, I project the integrated dataset into two dimensions via Principal Component Analysis and reveal latent groups with k -means; in this way I was able to obtain clustering visualization allowing users to focus onto clusters of countries and through this approach user can find and examine them to explore common trends and also discover by exploration new epidemiological and diagnostic hypotheses reasoning over all linked views.

This tool transforms static MS statistics into a living, interactive solution where clinicians, epidemiologists, health authorities or simply the patients themselves can pose questions, discover patterns, and generate hypotheses in a user-friendly way.

II. MULTIPLE SCLEROSIS

Multiple Sclerosis (MS) is a chronic, often disabling disease of the central nervous system, which includes the brain, spinal cord, and optic nerves. It occurs when the immune system mistakenly attacks the protective layer (myelin) that surrounds nerve fibers, leading to inflammation and disrupted communication between the brain and the rest of the body. Over time, this can cause permanent nerve damage.

MS affects each individual differently, but common symptoms include fatigue, difficulty walking, numbness or tingling, muscle weakness, vision problems, and problems with coordination or balance. Some people may experience long periods of stability, while others may see a steady progression of symptoms. The disease typically appears between the ages of 20 and 40, and it affects more women than men.

There are several types of MS, with the most common being relapsing-remitting MS (RRMS), which is characterized by episodes of new or worsening symptoms (relapses) followed by periods of partial or complete recovery (remission). Other types include secondary progressive MS (SPMS), primary progressive MS (PPMS), and these progressive forms are characterized by a continuous worsening of the disease.

While the exact cause of MS remains unknown, it is believed to be triggered by a combination of genetic and environmental factors. Although there is no cure, a variety of treatments exist to manage symptoms, reduce relapses, and slow the progression of the disease. Advances in medical research continue to improve our understanding and treatment options for MS.

III. RELATED WORK

This project was inspired by two important resources that explore the epidemiology of Multiple Sclerosis.

Atlas of MS

The *Atlas of MS*, maintained by the Multiple Sclerosis International Federation (MSIF), was the starting point for my idea. It offers an interactive map that lets users explore country-level MS data (such as the number of people living with MS or incidence rates) through a simple choropleth map view.

The aspect that struck me the most was how accessible this interface made the global MS burden. However, I also noticed some limitations: users can only view one metric at a time, there's no easy way to compare across years, and the tool doesn't support cross-indicator filtering or multivariate exploration.

AISM – Barometro della Sclerosi Multipla

On a national level, the Italian Multiple Sclerosis Association (AISM) publishes the *Barometro della Sclerosi Multipla*, a detailed yearly report that covers MS incidence, prevalence, healthcare services, and socio-economic data for every region in Italy. Although the Barometro is not interactive, it provides a comprehensive view of the disease in the Italian context and underlines the importance of combining clinical data with social and territorial indicators.

This crucial idea on integrating epidemiological data with contextual and social informations has also shaped my work (especially the decision to include risk factors like smoking) in the visual analytics application.

Together, the Atlas of MS and the AISM Barometro represent two valuable perspectives (global and national perspectives) that allowed me to define the structure, goals, and content of my project.

IV. DATASETS

In this section I'm going to describe where and how data of this project have been obtained. I have decided to develop and use a data pipeline in which first I have re-engineered data about MS by cleaning raw dataset and highlighting specific data from selected metrics, saving these final results into 3 CSV files, one for each analysis year, subsequently I've decided to enrich MS data with global smokers dataset from GBD. The outcome is a data source tracking World countries across three time periods of deep analysis merging informations about MS and smokers data, ready for visual data exploration, with the main goal of obtaining a reliable set of informations that can be used and implemented into this developed project of visual analytics.

A. Atlas of MS

The *Atlas of MS* is the most complete global resource with focus on the epidemiology of Multiple Sclerosis to date, developed through an extensive international collaboration, where instead of relying on a single centralized source, data

were gathered from a diverse network of national experts, neurologists, ministries of health, and MS organizations across multiple countries. Initiated by the Multiple Sclerosis International Federation (MSIF), this third edition gathered data from 115 countries, covering 87% of the world's population. Subjects involved in data collection have been able to collect informations about prevalence, incidence, diagnosis age, MS types, and healthcare system capacity. A main feature that characterizes this dataset is: Atlas directly engages national experts to report current data and provides full public access to survey responses, supporting awareness and policy change initiatives through the use of verified data and statistics.

Usable dataset is distributed as a downloadable Excel file (.xlsx) where user can find three sheets labeled 2008, 2013, and 2020–2022. Each worksheet contains 115 rows (corresponding to the reporting countries or territories) and approximately 40 columns spanning five thematic domains: (i) core epidemiology (prevalence and incidence rates), (ii) MS disease types, (iii) diagnostic timing and delay, (iv) access to treatment, and (v) disability metrics based on EDSS. Data are expressed as a mix of raw counts, rates per 100,000 inhabitants, and proportions.

For this project of Visual Analytics I have preferred to focus on a specific subset of metrics, retrieved from previously mentioned dataset, to emphasize the dataset's epidemiological focus and enable the discovery of meaningful patterns across countries and indicators. Selected and used metrics are:

- Total Number of People with MS
- Number of Prevalent Cases
- Prevalence of MS per 100,000
- Incidence of MS per 100,000
- Mean Age of Onset
- Numbers of MS Types % RRMS, PPMS, SPMS, Unknown
- World Regions
- Countries

For previously mentioned purpose I have preferred to restructure the column related to analyzed global countries such that I will be able to do a join with list of countries predefined in the D3 library's map data with respect to names. I have also added a column indicating the corresponding WHO geographical macro-region (i.e. macroregions of the World Health Organization) at continent level, this allowed me to establish a clear mapping between country and its related continent of belonging.

To avoid unnecessary computations, backend architecture calculates a SHA-256 hash for each generated file and compares it to a dictionary of expected hashes. Reprocessing is triggered only when the hash of the source workbook has changed, indicating that the data has been modified.

B. GBD 2019: Smokers Dataset

An emerging neurological research field highlights that heavy smoking aggravates MS progression. The smoking dataset that I have chosen to use is the Global Burden of Disease (GBD) 2019 database by IHME Organization and

provides estimates of the number of smokers across countries and regions, grouped by sex and year. The GBD 2019 Smoking Tobacco Use Prevalence dataset, gathered by the Institute for Health Metrics and Evaluation (IHME), provides estimates of smoking tobacco use across global countries and territories from 1990 to 2019. It includes data on the prevalence of smoking, the number of current users of smoked tobacco products, and supply-side tobacco availability and consumption. These estimates are disaggregated by sex, age group, and year, providing valuable insights into global smoking trends and their associated health problems.

V. VISUALIZATIONS

The visualization system developed in this project adopts a coordinated multiple views approach to support the exploration of complex epidemiological data. All visual components are based on a consistent, structured country- or region- dataset (with respect to year selection), allowing synchronized updates across views when the selected year changes. The choice of visualization techniques (including choropleth map, bubble chart, parallel coordinates, and clustered PCA plots) was guided by the need to capture geospatial distribution, inter-variable relationships, temporal comparisons, and hidden discoverable patterns among metrics and subjects analyzed. Main coordinated interaction has been implemented with respect to year selection, this choice allows a comparative analysis across time while maintaining a general perspective of MS epidemiology context.

A. Choropleth Map

One of the central views of the system is a choropleth map visualized using a 2-dimensional projection to understand the global distribution of MS cases. Users can select between two epidemiological indicators (prevalence and incidence) each inferred across three available reference years (2008, 2013, and 2020–2022). The selected measure indicator determines the map's color encoding (i.e. shades of blue for prevalence distributions while the red ones for incidence cases), each country will be colored with a specific red or blue grade such that user can understand numerical value of selected metric with respect to time period, this mechanism is supported by using a legend that allows understanding of the numerical value corresponding to the color hue. When a valid value is present, each country is shaded accordingly; missing data are rendered using a black–white striped pattern to give an idea of the lack of data for that specific country. Tooltip interactivity on mouse hover reveals the exact metric value for each country, aiding immediate visual interpretation. The color scale dynamically adjusts based on the data distribution for the selected year and measure, allowing for consistent cross-country comparison.

B. Bubble Chart

By implementing a bubble chart I wanted to have the possibility to explore the relationship between smoking habits and the prevalence of multiple sclerosis (MS) across global

regions. This deep analysis is conducted for each possible selectable dataset year (2008, 2013, and 2020–2022). In this visualization the x-axis represents the number of smokers, while the y-axis shows the number of MS cases while the bubble size encodes the ratio of MS cases to the number of smokers: in this way users can understand where elevated number of MS cases appears disproportionately high relative to smoking prevalence in a World region. To design this chart, I decided to merge the two datasets: the one on MS epidemiological data and the one on smokers globally, with the peculiarity of including entries for smokers only with value “Both” for the attribute “Sex” (`sex_name = "Both"`). Regional coloring and hover tooltips further enhance interpretability, such that users can have a quick comparison across continents and identification of notable outliers or trends.

C. Stacked Bar Chart

In order to capture the clinical heterogeneity of multiple sclerosis various forms, I decided to use and implement a stacked bar chart that allows user to visualize the distributions of disease-course types (i.e. Relapsing-Remitting (RRMS), Primary Progressive (PPMS), Secondary Progressive (SPMS), and Unknown (UCS)) within each WHO macro-region. One important note related to analyzed years, for this visualization chart it is possible to select only time intervals (2013 and 2020–2022): this due to the fact dataset about MS data doesn't contain informations about various MS forms for 2008 year. For each year selected by the user (2013 or 2020–2022), country-level data are aggregated to regional average, representing these elaborated data, user can visualize comparisons across large geographic areas. The stacked format highlights variation in case composition, making it easy to spot regional disparities in progressive forms of the disease. Hovering over each segment reveals the mean percentage cases for that subtype within the selected WHO macro-region. The chart provides in graphic form an overview of disease-course proportions expressed as percentages, in this way, the end user of this Visual Analytics application can extract possible inferences and use percentages of MS types that can support further investigations of access to care and health care planning to improve the distribution of medicines and therapies relative to the type percentages obtained in the visual analysis using Stacked Bar.

D. Parallel Coordinates Chart

The parallel coordinates chart offers a multivariate view for a selected set of key MS indicators related to WHO macro-regions: by implementing it I was able to obtain a comparison of trends over time. For each region, it's possible to choose among three quantitative dimensions that are: Number of Prevalent Cases, Mean Age of Onset, and Prevalence of MS per 100,000 and all of them are represented across three reference years (2008, 2013, and 2020–2022), such that it's visualized quantitative trend changes with respect to these moments of time. Each region is represented by a line connecting its values across these years, each color encodes a different

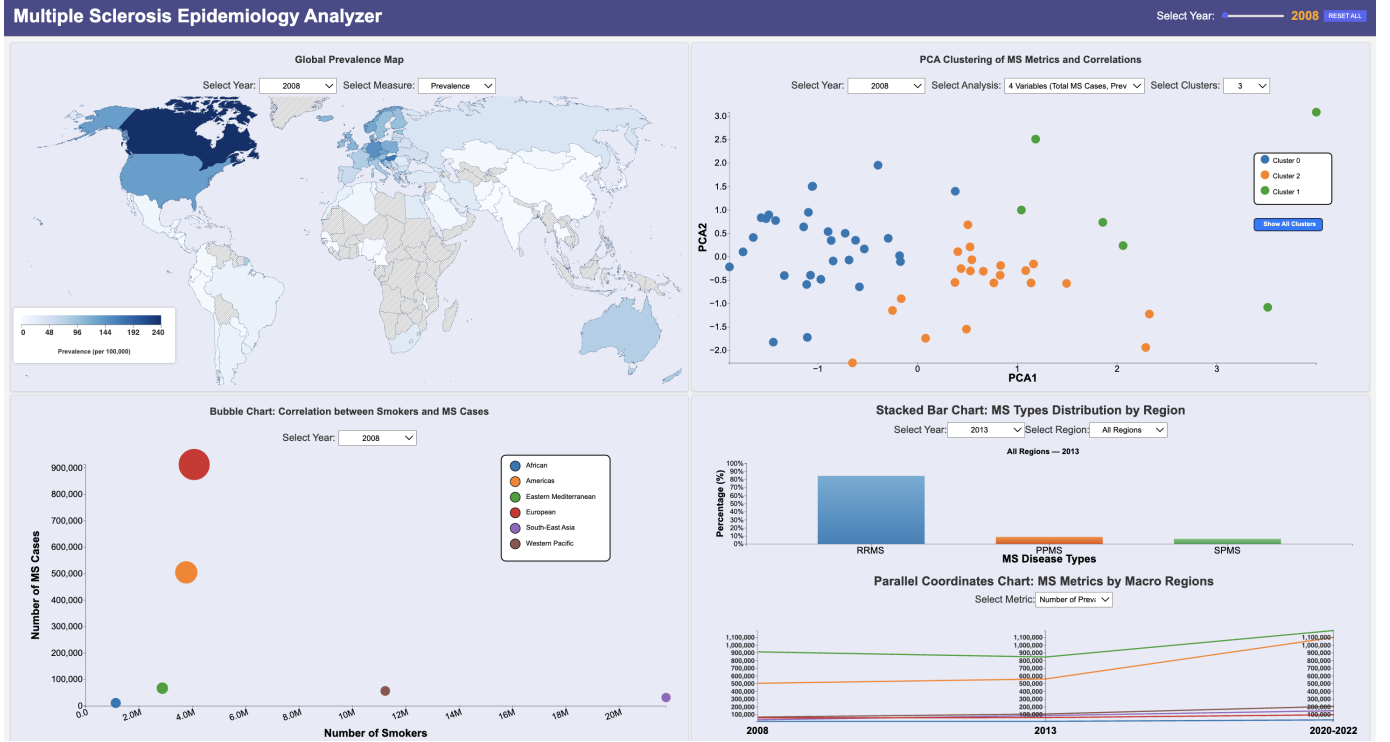


Fig. 1. Multiple Sclerosis Epidemiology Analyzer

WHO macro-region. The chart is particularly effective in revealing broad temporal trends, similar trends among regions over time, or rapid changes about MS selected metric across regions: with this implemented visualization it's offered to users a compact yet informative summary of regional evolution over time.

E. PCA Scatter Plot

To uncover the hidden structures and dynamics in epidemiological data, I decided to implement in this project a scatter plot based on Principal Component Analysis (PCA). The dimensionality of selected MS-related variables is reduced to two orthogonal axes, allowing for visual inspection of country-level similarities and trends. The PCA is applied to different combinations of indicators, including two-variable (i.e. Total MS cases and Mean Age of Onset), four-variable (adding Prevalence and Incidence to previous two variables case), and finally a data merge between smokers data and MS cases, and is computed independently for each year of interest (2008, 2013, and 2020–2022). A k -means clustering algorithm is then applied to the resulting PCA space, I also have included the ability to be able to select the number of clusters (k) by the user. Points are colored by cluster, and hover interactions reveal country- and region-level details. This visualization chart supports the identification of distinct epidemiological profiles and allows comparison of clustering patterns across different dimensions and time periods. Finally, the user is able to select the specific component elements of

the cluster by simply clicking on the cluster name in the legend related to the chart.

VI. ANALYTICS

The visual analytics application relies on a lightweight Python backend to prepare, process, and serve all the data required for interactive exploration. Each result is returned in JSON format, tailored for immediate integration into the frontend charts.

Key processing steps write their outputs to a specific folder on Pythonanywhere cloud as CSV files and track changes using SHA-256 hashes, ensuring that data is only recomputed when necessary. This setup supports fast, consistent visual feedback while keeping computational costs under control.

A. Data Extraction (/api/runDataExtraction)

This endpoint kicks off a simple ETL process (short for Extract, Transform, Load) used to get the data into shape for analysis and visualization. When a POST request is made, the backend loads the Atlas workbook using `pandas.read_excel`, extracts the main epidemiological variables (as described in Section III), and cleans up the `Country` field by trimming whitespace. It then saves three cleaned and structured CSV files (`Selected_Data_2008.csv`, `Selected_Data_2013.csv`, and `Selected_Data_2020-2022.csv`) each corresponding to a different reference year. Missing values are preserved in this phase to keep the visualizations complete, but they're

filtered out later when they might interfere with computations like standardization or PCA.

B. Caching and Integrity Control Mechanism

To avoid redundant computation and ensure data integrity across executions, the backend implements a lightweight caching mechanism based on SHA-256 file hashing. This strategy is encapsulated in three core functions: `compute_file_hash()`, `needs_update()`, and the global `EXPECTED_HASHES` dictionary.

a) Hash Computation (`compute_file_hash()`):

This function takes a file path and returns its SHA-256 hash by reading the file in binary mode and incrementally updating a hash object. It ensures that the computed hash reflects the exact binary content of the file.

b) Cache Validation (`needs_update()`):

Before regenerating any result file, the backend invokes `needs_update()` to determine whether recomputation is necessary. The function compares the current hash of the target file with its expected hash from the `EXPECTED_HASHES` dictionary. If the file does not exist or the hashes mismatch, the function returns `True`, signaling that the file must be updated.

c) Hash Tracking (`EXPECTED_HASHES`):

The `EXPECTED_HASHES` dictionary stores expected SHA-256 hashes for each result file, keyed by filename. These values are initialized at startup and updated dynamically whenever a file is recomputed. By maintaining this mapping in memory, the backend ensures that all preprocessing routines are executed only when their input data has changed.

d) Integration in Data Pipelines:

All PCA and data extraction routines (e.g., `runPCA2_all()`, `runMergePCA_all()`, `runDataExtraction()`) integrate this logic. Before writing output, they first check whether recomputation is necessary. Once a new file is saved, its hash is computed and stored in `EXPECTED_HASHES` for future validation.

This mechanism achieves three goals: (i) reducing computational load, (ii) ensuring that no stale data is reused, and (iii) supporting reproducibility by verifying that results match their known-good hash.

C. Batch PCA Clustering Endpoints

Each of these three endpoints performs PCA (Principal Component Analysis) and clustering over multiple years (2008, 2013, and 2020–2022), generating a complete set of output files for frontend visualization and interaction.

`/api/runPCA2_all` performs PCA on two epidemiological variables: total number of people with MS and mean age of onset. The data is first standardized using `StandardScaler`, and then PCA is applied to reduce it to two principal components. Clustering is performed using `KMeans` for values of k ranging from 3 to 6, with each result saved to a separate CSV file that includes the PCA coordinates and cluster assignments.

`/api/runPCA4_all` uses a four-dimensional feature set consisting of: total number of people with MS, prevalence per 100,000, incidence per 100,000, and mean age of onset. The pipeline follows same idea of previous case: standardization, PCA to two components, clustering for $k = 3$ to 6, and CSV export for each configuration.

`/api/runMergePCA_all` extends this pipeline by merging MS epidemiological data with smoking prevalence estimates. For each year, the system filters smoking data for the total population (`sex_name = "Both"`) and joins it with MS data by country. PCA is then applied to two features: number of smokers and number of people with MS. The resulting dataset is clustered and saved for multiple k values. This merged view enables cross-indicator analysis of potential correlations between smoking habits and MS prevalence.

All three endpoints implement a caching strategy based on SHA-256 file hashes. Before computing any result, the system checks whether a matching file already exists and whether its contents are unchanged. If so, computation is skipped. This mechanism ensures that expensive operations are only executed when necessary and that all results remain reproducible over time.

D. PCA Result Retrieval Endpoints

These three endpoints serve as lightweight access points for retrieving PCA and clustering results previously computed using the corresponding `_all` batch endpoints. They are typically called by the frontend application in response to user input, such as selecting a year or cluster count k in an interactive visualization.

`/api/runMergePCA` extracts PCA clustering results from the merged MS and smoking dataset. The client must provide a JSON payload specifying both the year (e.g., "2013") and the number of clusters (e.g., "k": 4). If the corresponding CSV file exists (created earlier by `/api/runMergePCA_all`) the server loads it, sanitizes the numeric fields via `clean_dataframe()`, and returns the data in JSON format. If the file is missing, a 404 error is returned with instructions to precompute it.

`/api/runPCA2` follows the same retrieval logic, but returns PCA clustering results based on two MS-related variables: total number of people with MS and mean age of onset. It assumes that the required output has been generated in advance using `/api/runPCA2_all`.

`/api/runPCA4` works similarly, but targets the more complex four-variable clustering model based on total MS cases, prevalence per 100,000, incidence per 100,000, and mean age of onset. It returns the corresponding PCA-reduced data and cluster labels if the requested file is available.

These endpoints allow the frontend to dynamically load just-in-time results without re-triggering expensive computations. They rely entirely on precomputed files, making the overall system responsive, modular, and efficient.

E. Retrieval of Data

Each visualization in the application relies on a specific API endpoint to retrieve the data it needs. These endpoints

return preprocessed data files or clustering results that are used to feed the charts. Each one serves a precise purpose in supporting interactive exploration of the MS dataset.

Choropleth Map: The choropleth map loads country-level MS indicators for a given year. The data includes numerical values such as prevalence or incidence, organized by country. This information is used to visually map the intensity of MS across the globe, allowing users to quickly spot geographic patterns.

Bubble Chart Plot: To investigate possible correlations between smoking and MS, the bubble chart plot retrieves two datasets: one reporting the number of smokers and the other detailing MS prevalence and totals. The data is merged by region and year, allowing the frontend to plot a joint view of lifestyle factors and disease impact.

Stacked Bar Chart: This endpoint returns the composition of MS disease types (RRMS, SPMS, PPMS, and unknown cases) aggregated by WHO macro-region. The data enables a side-by-side comparison of disease-course heterogeneity, helping identify regions where progressive forms of MS are more common.

Parallel Coordinates Chart: For this visualization, the data includes several MS-related indicators grouped by region. Metrics such as total cases, mean onset age, prevalence, and incidence are bundled per year and macro-region. The chart uses these variables to allow comparative multivariate analysis across different parts of the world.

PCA – Two Variables: The PCA endpoint for two variables serves precomputed clustering results based on total MS cases and mean age of onset. Users select the reference year and number of clusters, and the backend returns a reduced version of the dataset where each country is mapped to two PCA components and a cluster label.

PCA – Four Variables: This variant of PCA uses four key indicators: total MS cases, prevalence per 100,000, incidence per 100,000, and mean age of onset. The resulting dataset helps identify more nuanced clustering patterns, where countries with similar MS profiles are grouped together.

PCA – MS and Smoking Combined: This endpoint retrieves clustering results based on a merged dataset combining MS statistics with smoking data. Each point in the dataset represents a country and includes values for number of smokers and total MS cases. PCA is applied to reveal latent correlations and segment the countries into behavioral or risk-based clusters.

All of these endpoints are designed to return structured, ready-to-visualize data in JSON format. They are called dynamically by the frontend whenever a user selects a chart, year, or clustering option. This setup keeps the interface responsive while ensuring consistency in the underlying data served.

VII. INSIGHTS

In this section I will analyze how the visual analytics tool enables users to explore data and uncover meaningful patterns. By integrating various visualizations users can ask themselves pertinent questions related to MS burden, care requirements, and potential risk factors. Below are four possible illustrative

use cases that exemplify interactions with the tool but, considering the application domain, there may be many others.

A. Scenario 1: Supporting Neurorehabilitation Planning through Regional MS Trends

A public health analyst uses the choropleth map to identify countries with a high overall number of MS cases, focus and reasoning also in terms of WHO regions. This spatial overview is complemented by the parallel coordinates chart, which reveals a steady increase in case numbers over time across several regions. By combining these views, the analyst can identify areas where the MS number of cases trend is not only high but also growing. These insights support evidence-based decisions about where to enhance neurorehabilitation structures and prepare for possible rising care demands in specific parts of the world.

B. Scenario 2: Smoking as a Modifiable Risk Factor

An epidemiologist explores the relationship between smoking and the MS burden across WHO regions. Using implemented bubble chart, it's possible to compare the number of smokers (x-axis) with the number of MS cases (y-axis), where bubble size reflects the ratio between these two numerics. Certain regions stand out with a strong correlation between high smoking rates and a greater number of MS cases. To further investigate, the analyst turns to the stacked bar chart and PCA clustering visualization about correlation between MS cases and smokers, such that it is possible to observe that these same regions also report higher proportions of more severe MS types. This pattern suggests a possible link between smoking prevalence and disease progression, offering interesting insight for targeted prevention and well-health education strategies against bad habits like smoking.

C. Scenario 3: Strategic Targeting of DMT Launches by WHO Region

A pharmaceutical strategist seeks to identify WHO regions where introducing new Disease-Modifying Therapies (DMTs) for Multiple Sclerosis (MS) would be most impactful. Utilizing the stacked bar chart, they observe that regions like the Americas and Europe consistently exhibit a high proportion of relapsing-remitting MS (RRMS) cases over time as also known in literature. Complementing this, the parallel coordinates chart reveals a concurrent rise in overall MS cases within these regions. This alignment suggests a growing demand for RRMS-targeted treatments. By integrating insights from both visualizations, the strategist can prioritize regions with increasing MS number cases and a significant RRMS population, informing strategic decisions on DMT deployment.

D. Scenario 4: Exploring Temporal Trends Across Years

A data analyst explores how MS indicators have evolved over time across different macro-regions. Employing the parallel coordinates chart, they analyze data from 2008, 2013, and 2020–2022, focusing on metrics like "Number of Prevalent Cases," "Prevalence per 100,000," and "Mean Age of Onset."

The visualization reveals trends such as increasing prevalence in Europe and the Americas, stable low levels in Africa (this trend also due to the low number of diagnoses carried out in this region). This temporal analysis helps to understand the epidemiological trends of MS at the regional level.

VIII. CONCLUSION

The visual-analytics application I have described in this report merges global epidemiological metrics from the *Atlas of MS* with lifestyle data from GBD 2019 about smoking habits, with the goal of translating a high-dimensional static spreadsheet dataset into an enriched interaction workspace that scales from world-wide overviews to country-level detail. By combining choropleth mapping, correlation bubble plots, phenotype-aware stacked bars, colour-encoded parallel coordinates and PCA-based clustering, the system could allow clinicians, researchers and policy makers to interrogate Multiple-Sclerosis cases data by using an interactive approach and to navigate instantly between spatial, temporal and multivariate perspectives. I want to implement and use inside my project an analytics pipeline exposing endpoints, ensuring an easy reproducibility and open re-use in other possible extension or advanced works.

Possible future developments that could be implemented include:

- I plan to integrate additional pathologies usually associated with MS (i.e. obesity, vitamin D deficiency, and physical inactivity). In this way user will be able to explore multi-factor risk patterns and better understand how lifestyle and environmental factors may influence disease distribution across regions.
- I plan to develop an analysis on how many people in each region have been exposed to EBV and link this to MS case numbers, so users can see where higher EBV exposure seems to coincide with more MS cases. Long-term studies show that the risk of developing multiple sclerosis rises by more than 30 times after EBV infection. This possible future implementation will help researchers and clinicians explore how viral exposure might spark or worsen MS progression.
- It could be introduced user-defined filtering and comparison features, allowing users to create custom country or region groups or variable combinations and compare them interactively. In such way users are able to investigate specific hypotheses or regional or country-level trends more deeply.
- A possible future work is to create a mixed-methods evaluation to assess how intuitive and effective the tool is in practice, such that I can directly understand the effectiveness of this application. Neurologists and public health experts could be invited to use the app, sending their feedbacks such that I will be able to improve this tool and insert new specific and requested features, solving related use problems.

REFERENCES

- [1] Multiple Sclerosis International Federation (MSIF), "Global MS statistics and resources," 2025. [Online]. Available: <https://www.msif.org/>
- [2] MSIF, "Atlas of MS — Global map of MS epidemiology," 2025. [Online]. Available: <https://www.atlasofms.org/map/global/epidemiology/number-of-people-with-ms>
- [3] MSIF, "Atlas of MS, 3rd Edition," September 2020. [Online]. Available: <https://www.aism.it/sites/default/files/Atlas%203rd%20Edition.pdf>
- [4] IHME, "Global Burden of Disease Study 2019 (GBD 2019) Smoking Tobacco Use Prevalence 1990–2019," Institute for Health Metrics and Evaluation, 2021. [Online]. Available: <https://ghdx.healthdata.org/record/ihme-data/gbd-2019-smoking-tobacco-use-prevalence-1990-2019>
- [5] S. Ralston, M. J. Rodgers, and L. Hawkins, "Geo-visual exploration of global multiple sclerosis prevalence," in *IEEE VIS Posters*, 2022.
- [6] Associazione Italiana Sclerosi Multipla (AISM), *Barometro della Sclerosi Multipla e patologie correlate 2023*. Genova, Italy: AISM, 2023. [Online]. Available: https://www.aism.it/sites/default/files/2023-10/barometro_2023_web.pdf
- [7] National Institute of Neurological Disorders and Stroke (NINDS), "Multiple Sclerosis Information Page," 2024. [Online]. Available: <https://www.ninds.nih.gov/health-information/disorders/multiple-sclerosis>
- [8] Bjornevik K., Münz C., Cohen J. I., Ascherio A., et al., "Epstein–Barr virus as a leading cause of multiple sclerosis: mechanisms and implications," *Nature Reviews Neurology*, vol. 19, pp. 160–171, 2023. [Online]. Available: <https://www.nature.com/articles/s41582-023-00775-5>
- [9] Leggett H., "Study identifies how Epstein-Barr virus triggers multiple sclerosis," *Stanford Medicine News Center*, Jan. 24, 2022. [Online]. Available: <https://med.stanford.edu/news/all-news/2022/01/epstein-barr-virus-multiple-sclerosis.html>
- [10] Soldan S. S. and Lieberman P. M., "Epstein-Barr virus and multiple sclerosis," *Nature Reviews Microbiology*, vol. 21, no. 1, pp. 51–64, Jan. 2023. [Online]. Available: <https://pubmed.ncbi.nlm.nih.gov/35931816/>